Chapter 5 exercises

Work over $\mathbb{R}$ unless specified otherwise. State the determinant property used in each argument.

CH05-E1

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix}.$$

1. Compute $\det A$ by expansion along row 1.
2. Add twice column 1 to column 2 to obtain B. Find $\det B$.
3. Explain why the determinants agree.

CH05-E2

1. Write the six signed products in the 3×3 permutation formula.
2. Identify the even and odd permutations.
3. Construct a nonzero 3×3 matrix with exactly two nonzero permutation products, and compute its determinant.

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CH05-E3

For

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix},$$

1. Find C_{11}, C_{12}, C_{13}.
2. Compute the determinant by expansion along row 1.
3. Repeat along row 2 and compare.

CH05-E4

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix}.$$

Let $C = [C_{ij}]$ be the cofactor matrix.

1. Find C^T and use the adjugate to compute A^{-1}.
2. Verify $A \operatorname{adj} A = (\det A) I$.
3. Explain why this solves $Ax = b$, and why it is unsuitable for a 1000×1000 numerical matrix.

CH05-E5

Solve

$$\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

1. Use Cramer's rule.
2. Verify by direct substitution.
3. Repeat for right-hand side $(2, 5)^T$.

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CH05-E6

1. Find the signed and unsigned parallelogram areas for $u = (2, 1)^T, v = (1, 3)^T$.
2. For $w = (u_1, u_2, u_3)^T, z = (v_1, v_2, v_3)^T$, prove their cross product is perpendicular to both.
3. For $w = (1, 0, 2)^T, z = (0, 1, -1)^T$, compute $w \times z$ and $\det[w\ z\ (w \times z)]$.

CH05-E7

1. Prove $\det(cA) = c^n \det A$ for $A \in \mathbb{R}^{n \times n}$.
2. Prove $|\det Q| = 1$ for real orthogonal Q.
3. Prove that a real skew-symmetric matrix of odd size is singular. Give a counterexample in even size.

CH05-E8

Let

$$A = \begin{bmatrix} a & b & 0 \\ c & d & 1 \\ 0 & 2 & a+d \end{bmatrix}.$$

1. Express $\det A$ in terms of a, b, c, d.
2. Choose parameters giving a nonzero singular matrix and explain its geometry.
3. For $a = d = 1, b = c = 0$, test invertibility before computing the characteristic polynomial. Relate the result to trace and determinant.

CH05-E9

Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, with

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad B = I_2, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}.$$

Find the Schur complement and $\det M$, showing the elimination matrix and its determinant. If A is singular, must M be singular? Give a counterexample.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 5.

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